

EGFR fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes EGFR gene fusions from the msk_impact_50k_2026 study, covering 55 fusion events. 28/55 fusions (50.9%) were in-frame. 41/55 fusions (74.5%) retained the Protein tyrosine and serine/threonine kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0114539$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for EGFR.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 55 analyzed fusion events, identifying 31 distinct fusion partner genes. The most frequent partner was SEPTIN14, observed in 18 events.

Domain retention

Protein tyrosine and serine/threonine kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 25/28 (89.3%) of in-frame fusions retained the domain, $p=0.0114539$ (statistically significant at $\alpha=0.05$). The Protein tyrosine and serine/threonine kinase appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.247525.

Domain disruption

Disruption of the Receptor L domain, Furin-like cysteine rich region, and Growth factor receptor domain IV was tested with Fisher's exact test comparing in-frame fusions against all others; 6/28 (21.4%) of in-frame fusions disrupted the domain, $p=0.991541$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.920792.

Cutpoint detection

Cutpoint detection scanned 55 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 982 aa (permutation-corrected $p=0.128713$, not statistically significant at $\alpha=0.05$). This is 17 aa from the nearest configured domain boundary at 965 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=41$) and "not_retained" ($n=14$). For Frame_status, retained: 25/31 (80.6%, 95% CI 63.7%-90.8%); not_retained: 3/9 (33.3%, 95% CI 12.1%-64.6%). Welch's t-test comparing Tumor_variant_count between groups gave means of 358.805 vs 214.071, $p=0.389355$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (31 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: EGFR has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SEPTIN14	18	0.32727272727272727	0.060638136511060495	0.0035838425228707435	0.9786998132586368	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32105576577621364	1
RAD51	2	0.03636363636363636	0.060638136511060495	0.0035838425228707435	0.9989495798319328	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23682050348948075	2
LANCL2	3	0.05454545454545454	0.060638136511060495	0.0035838425228707435	0.9618347338935574	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2367078220532699	3
ELAPOR2	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.9989495798319328	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23136595803493532	4
PKD1L1	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.9989495798319328	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23136595803493532	5
VPS41	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.9957983193277311	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23089326895930506	6
ZNF713	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.9401260504201681	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2225424286231706	7
VWC2	2	0.03636363636363636	0.060638136511060495	0.0035838425228707435	0.9028361344537815	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22240348668275806	8
NIPSNAP2	3	0.05454545454545454	0.060638136511060495	0.0035838425228707435	0.8658963585434174	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22231706575074892	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TNRC18	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.9233193277310925	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22002142021980928	10
BLTP3B	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.9023109243697479	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.21687015971560758	11
CDH7	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.832983193277311	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20647100005174204	12
FADD	2	0.03636363636363636	0.060638136511060495	0.0035838425228707435	0.6901260504201681	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19049697407771604	13
EEA1	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.6911764705882353	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1851999916483807	14
KIF5B	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.6911764705882353	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1851999916483807	15
VSTM2A	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.6911764705882353	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1851999916483807	16
CADPS	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.6901260504201681	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1850424286231706	17
NUMA1	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.6901260504201681	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1850424286231706	18
PDE1C	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.6523109243697479	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17937015971560757	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
YIF1B	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.6407563025210083	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17763696643829666	20
TUT7	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.6271008403361344	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17558864711056557	21
CSF2RA	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.3214285714285714	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1297378067744311	22
DDC	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.2804621848739496	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12359284879123783	23
GARS1	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.2804621848739496	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12359284879123783	24
SEL1L	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.2804621848739496	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12359284879123783	25
TNS3	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.2804621848739496	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12359284879123783	26
PCDH15	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.2710084033613446	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12217478156434708	27
VOPP1	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.23004201680672265	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1160298235811538	28
SCAF4	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.11764705882352944	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.09917057988367481	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ZBPB	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.006302521008403339	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.0824688992114059	30
PLOD3	1	0.01818181818181818	0.060638136511060495	0.0035838425228707435	0.0	0.6019267297326067	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.0815235210601454	31

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
30	1	40	0	14	None	1.0	-0.0
36	2	39	0	14	None	1.0	-0.0
142	3	38	0	14	None	0.5624166190203925	0.24994185469837954
249	3	38	1	13	1.0263157894736843	1.0	-0.0
288	3	38	2	12	0.47368421052631576	0.5924468510484049	0.22735060452753406
297	7	34	2	12	1.2352941176470589	1.0	-0.0
336	8	33	2	12	1.4545454545454546	1.0	-0.0
599	9	32	2	12	1.6875	0.7093311767774617	0.14915095146564436
627	9	32	3	11	1.03125	1.0	-0.0
640	10	31	3	11	1.1827956989247312	1.0	-0.0
651	11	30	3	11	1.3444444444444446	1.0	-0.0
687	13	28	5	9	0.8357142857142857	1.0	-0.0
688	15	26	6	8	0.7692307692307693	0.7548366596998944	0.12214701587495107
823	15	26	7	7	0.5769230769230769	0.528760430668828	0.27674105235972185
875	15	26	9	5	0.32051282051282054	0.1175979559898796	0.9296002268145651
981	15	26	10	4	0.23076923076923078	0.031971018585759146	1.4952435270336717
982	21	20	12	2	0.175	0.02895546314423952	1.5382694841107174
983	31	10	13	1	0.23846153846153847	0.25480492869908067	0.5937921756847577
986	32	9	13	1	0.27350427350427353	0.4229368950729017	0.37372442734387856
994	33	8	13	1	0.3173076923076923	0.4208384837822578	0.3758845527657329
1003	34	7	13	1	0.37362637362637363	0.663943317750199	0.17786899569082928
1039	35	6	13	1	0.44871794871794873	0.663943317750199	0.17786899569082928

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1054	37	4	13	1	0.7115384615384616	1.0	-0.0
1055	38	3	14	0	0.0	0.5624166190203926	0.24994185469837946
1060	39	2	14	0	0.0	1.0	-0.0
1075	40	1	14	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

6	13
22	14

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf01030	Receptor L domain	55	36	1	18	0.012328117591275486
auto_disruption_pf00757	Furin-like cysteine rich region	55	35	7	13	0.1103896103896104
auto_disruption_pf14843	Growth factor receptor domain IV	55	39	2	14	0.07623106060606061

domain_retention tables

frame_domain_contingency_table

25	16
3	11

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	55	41	3	11	0.10304912478825523

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
18	43	55	0.7818181818181819

frequency tables

Partner_gene_counts

Partner_gene	Event_count
BLTP3B	1
CADPS	1
CDH7	1
CSF2RA	1
DDC	1
EEA1	1
ELAPOR2	1
FADD	2
GARS1	1
KIF5B	1
LANCL2	3
NIPSNAP2	3
NUMA1	1
PCDH15	1
PDE1C	1
PKD1L1	1
PLOD3	1
RAD51	2
SCAF4	1
SEL1L	1
SEPTIN14	18
TNRC18	1
TNS3	1
TUT7	1
VOPP1	1
VPS41	1
VSTM2A	1
VWC2	2
YIF1B	1
ZNF713	1
ZBPB	1

Per-event results (results.tsv)

event_id	sampl_e_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 1404-T01-IM6-2	P-0031 404-T01-IM6		BLTP3B-EGFR	BLTP3B	unknown	five_prime	55270272	27	1075	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - UHRF1BP1L (NM_015054) Rearrangement :t(7;12)(p11.2;q23.1)(chr7:g.55270272::chr12:g.100460508) Protein Fusion: mid-exon {EGFR:UHRF1BP1L}	NA
EVT-P-005 0679-T02-IM6-3	P-0050 679-T02-IM6		CADPS-EGFR	CADPS	out-of-frame	three_prime	55241099	17	687	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		CADPS (NM_003716) - EGFR (NM_005228) fusion: t(3;7)(p14.2;p11.2)(chr3:g.62537410::chr7:g.55241099) Protein Fusion: out of frame {CADPS:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 8683-T01-IM6-4	P-0038 683-T01-IM6		CSF2RA-EGFR	CSF2RA	unknown	threep_rime	55224220	9	336	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Growth factor receptor domain IV	Receptor domain	Furin-like cysteine rich region	CSF2RA (NM_001161530) - EGFR (NM_005228) rearrangement: t(7;X)(p11.2;p22.33)(chr7:g.55224220::chrX:g.1401662) Protein Fusion: mid-exon {CSF2RA:EGFR}	NA
EVT-P-001 6795-T01-IM6-5	P-0016 795-T01-IM6		DDC-EGFR	DDC	out-of-frame	threep_rime	55222204	7	297	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Growth factor receptor domain IV	Receptor domain	Furin-like cysteine rich region	DDC (NM_000790) - EGFR (NM_005228) Rearrangement : c.435+4059 :DDC_c.889+359:EGFRinv Protein Fusion: out of frame {DDC:EGFR}	NA
EVT-P-001 8389-T02-IM6-6	P-0018 389-T02-IM6		EEA1-EGFR	EEA1	in-frame	threep_rime	55241575	18	688	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor domain; Furin-like cysteine rich region ; Growth factor receptor domain IV		EEA1 (NM_003566) - EGFR (NM_005228) rearrangement: t(7;12)(p11.2;q22)(chr7:g.55241575::chr12:g.932211089) Protein Fusion: in frame {EEA1:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 4660-T01-IM7-9	P-0064 660-T01-IM7		EGFR-CDH7	CDH7	out-of-frame	five_prime	55254206	20	823	True	disrupted	0.43873517786561267	True	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		Protein tyrosine and serine/threonine kinase	EGFR (NM_005228) - CDH7 (NM_033646) rearrangement: t(7;18)(p11.2;q22.1)(chr7:g.55254206::chr18:g.63486938) Protein Fusion: out of frame {EGFR:CDH7}	NA
EVT-P-003 4892-T01-IM6-497	P-0034 892-T01-IM6		EGFR-ELAPOR2	ELAPOR2	unknown	three_prime	55268103	24	981	False	lost	0.0	False			Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	EGFR (NM_005228) rearrangement: c.2943:EGFR_chr7:g.86587163del Antisense Fusion	NA
EVT-P-005 9222-T03-IM7-500	P-0059 222-T03-IM7		EGFR-LANCL2	LANCL2	in-frame	five_prime	55259704	21	875	True	disrupted	0.6442687747035574	True	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		Protein tyrosine and serine/threonine kinase	EGFR (NM_005228) - LANCL2 (NM_018697) rearrangement: c.2625+137:EGFR_c.1008+3812:LANCL2del Protein Fusion: in frame {EGFR:LANCL2}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-005 7880-T01-I M6-501	P-0057 880-T0 1-IM6		EGFR -LAN CL2	LAN CL2	in-fr ame	five _pr ime	552688 71	25	983	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h facto r recepto r doma in IV			EGFR (NM_005228) - LANCL2 (NM_018697) fusion: c.2947-10:EGFR_c.2 05-8690:LANCL2del Protein Fusion: in frame {EGFR:LANCL2}	NA
EVT-P-006 3142-T01-I M7-503	P-0063 142-T0 1-IM7		EGFR -PCD H15	PC DH1 5	unk now n	five _pr ime	552218 20	7	288	False	lost	0.0	False	Recep tor L d omain	Prote in tyro sine and s erine /thre onine kinas e; Gr owth facto r rec eptor dom ain IV	Furin-li ke cyst eine rich region	EGFR (NM_005228) - PCDH15 (NM_001142763) rearrangement: t(7;10)(p11.2 ;q21.1)(chr7:g.55221820::chr 10:g.56187463) Protein Fusion: mid-exon {EGFR:PCDH15}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-004 2236-T01-I M6-504	P-0042 236-T0 1-IM6		EGFR -PKD 1L1	PKD 1L1	in-fr ame	five _pr ime	552688 24	25	983	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - PKD1L1 (NM_138295) fusion: c.2947-57:EGFR_c.8 527-7825:PKD1L1inv Protein Fusion: in frame {EGFR:PKD1L1}	NA
EVT-P-004 1149-T04-I M6-505	P-0041 149-T0 4-IM6		EGFR -RAD 51	RA D51	in-fr ame	five _pr ime	552687 99	25	983	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.5 5268799::chr15:g.40996189) Protein Fusion: in frame {EGFR:RAD51}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 1149-T01-IM6-506	P-0041 149-T01-IM6		EGFR-RAD51	RAD51	in-frame	five_prime	55268799	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.55268799::chr15:g.40996189) Protein Fusion: in frame {EGFR:RAD51}	NA
EVT-P-003 4594-T01-IM6-507	P-0034 594-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268544	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion : c.2947-110:EGFR_c.1120-1204:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038138-T01-IM6-508	P-0038138-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268381	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+275:EGFR_c.1119+2043:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0037526-T01-IM6-509	P-0037526-T01-IM6		EGFR-SEPTIN14	SEPTIN14	unknown	five_prime	55268916	25	994	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2982:EGFR_c.1119+1151:SEPT14inv Protein Fusion: mid-exon {EGFR:SEPT14}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-003 9255-T02-I M6-510	P-0039 255-T0 2-IM6		EGFR -SEP TIN14	SEP TIN 14	in-fr ame	five _pr ime	552708 07	27	1091	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3271+489:EGFR_c. 818-3507:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-003 2604-T01-I M6-512	P-0032 604-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	in-fr ame	five _pr ime	552687 71	25	983	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion : c.2947-110:EGFR_c.1120- 1204:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 1753-T01-IM6-513	P-0031 753-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268412	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+306:EGFR_c.1119+2531:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-000 5701-T01-IM5-514	P-0005 701-T01-IM5		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55269490	26	1054	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15:EGFR_c.1119+2833:SEPT14inv Protein fusion: in frame (EGFR-SEPT14)	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-004 3525-T01-I M6-515	P-0043 525-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	in-fr ame	five _pr ime	552688 59	25	983	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1 120-807:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-002 9159-T01-I M6-516	P-0029 159-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	in-fr ame	thre e_ pr ime	552698 50	27	1055	True	lost	0.0	False		Prote in tyr osine and s erine /thre onine kinas e; Re cepto r L d omai n; Fu rin-lik e cys teine rich r egion ; Gro wth f actor rece ptor dom ain IV		SEPT14 (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369 :SEPT14_c.3163-360:EGFRi nv Protein Fusion: in frame {SEPT14:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028241-T01-IM6-517	P-0028241-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268525	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2947-356:EGFR_c.1119+4312:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0025191-T02-IM6-518	P-0025191-T02-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55269542	26	1054	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3162+67:EGFR_c.1120-3396:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-006 1245-T01-I M7-519	P-0061 245-T0 1-IM7		EGFR -SEP TIN14	SEP TIN 14	in-fr ame	five _pr ime	552688 15	25	983	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement: c.2947-66:E GFR_c.1119+2973:SEPT14i nv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-001 4220-T02-I M6-520	P-0014 220-T0 2-IM6		EGFR -SEP TIN14	SEP TIN 14	unk now n	five _pr ime	552689 42	25	1003	False	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-25 fused with SEPT14 exon 10); c.300 8:EGFR_c.1120-4102:SEPT 14inv Protein Fusion: mid-exon {EGFR:SEPT14}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-002 3043-T01-I M6-521	P-0023 043-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	in-fr ame	five _pr ime	552681 79	24	982	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion (EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+73:EGFR_c.1120-1 180:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-001 9603-T01-I M6-522	P-0019 603-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	in-fr ame	five _pr ime	552681 71	24	982	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion ((EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+65:EGFR_c.1119+3 306:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_na me	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-004 7356-T01-I M6-523	P-0047 356-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	in-fr ame	five _pr ime	552683 16	24	982	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2946+210:EGFR_c. 1120-2817:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-001 6256-T01-I M6-524	P-0016 256-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	in-fr ame	five _pr ime	552683 90	24	982	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion(EGFR exons 1-24 fused to SEPT14 exons 9-10): c.2946+284:EGFR_c. 1120-3940:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015179-T01-IM6-525	P-0015179-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268650	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_002529) - SEPT14 (NM_014857) fusion (EGFR exons 1-24 fused with SEPT14 exon 10): c.2947-231:EGFR_c.1119+3910:SEPT14inv Protein fusion: in frame (EGFR-SEPT14)	NA
EVT-P-0058375-T01-IM6-526	P-0058375-T01-IM6		EGFR-TNRC18	TNRC18	out-of-frame	five_prime	55269897	27	1055	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - TNRC18 (NM_001080495) fusion: c.3163-313:EGFR_c.5720-6144:TNRC18inv Protein Fusion: out of frame {EGFR:TNRC18}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016009-T01-IM6-527	P-0016009-T01-IM6		EGFR-VOPP1	VOPP1	in-frame	five_prime	55220393	6	249	True	lost	0.0	False	Receptor L domain	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Furin-like cysteine rich region	EGFR (NM_005228) - VOPP1 (NM_030796): c.747+36:EGFR_c.55-23161VOPP1inv Protein Fusion: in frame {EGFR:VOPP1}	NA
EVT-P-0039262-T01-IM6-529	P-0039262-T01-IM6		EGFR-VSTM2A	VSTM2A	out-of-frame	five_prime	55241509	18	688	True	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - VSTM2A (NM_182546) rearrangement: c.2062-105:EGFR_c.634+2190:VSTM2A dup Protein Fusion: out of frame {EGFR:VSTM2A}	NA
EVT-P-0042236-T01-IM6-530	P-0042236-T01-IM6		EGFR-VWC2	VWC2	unknown	five_prime	55270225	27	1060	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - VWC2 (NM_198570) fusion: c.3178:EGFR_c.827-13914:VWC2dup Protein Fusion: mid-exon {EGFR:VWC2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043097-T01-IM6-531	P-0043097-T01-IM6		EGFR-ZNF713	ZNF713	in-frame	five_prime	55269400	26	1039	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - ZNF713 (NM_182633) fusion: c.3115-28:EGFR_c.49-962:ZNF713del Protein Fusion: in frame {EGFR:ZNF713}	NA
EVT-P-0027944-T01-IM6-534	P-0027944-T01-IM6		FADD-EGFR	FADD	unknown	five_prime	55241077	17	687	True	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330) Protein Fusion: mid-exon {EGFR:FADD}	NA
EVT-P-0027944-T03-IM6-535	P-0027944-T03-IM6		FADD-EGFR	FADD	unknown	five_prime	55241077	17	687	True	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330) Protein Fusion: mid-exon {EGFR:FADD}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_n_cated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 3743-T01-I M7-536	P-0063 743-T0 1-IM7		GARS 1-EG FR	GA RS1	out- of-fr ame	thr_ee _pr im_e	552223 68	7	297	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Grow th fac tor r ecepto r doma in IV	Rece ptor L do main	Furin-li ke cyst eine rich region	GARS (NM_002047) - EGFR (NM_005228) fusion: c.1032-338:GARS_c.889+523:EGFRdel Protein Fusion: out of frame {GARS:EGFR}	NA
EVT-P-002 1038-T01-I M6-538	P-0021 038-T0 1-IM6		KIF5B -EGF R	KIF 5B	in-fr ame	thr_ee _pr im_e	552414 86	18	688	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase	Rece ptor L do main; Furin- like cyste ine rich region ; Gro wth f actor rece ptor dom ain IV		KIF5B (NM_004521) - EGFR (NM_005228) rearrangement: t(7;10)(p11.2;q11.22)(chr7:g.55241486::chr10:g.32312089) Protein Fusion: in frame {KIF5B:EGFR}	NA
EVT-P-002 5018-T01-I M6-539	P-0025 018-T0 1-IM6		LANC L2-E GFR	LAN CL2	out- of-fr ame	thr_ee _pr im_e	552685 37	25	983	True	lost	0.0	False		Prote in tyr osine and s erine /thre onine kinas e; Re cepto r L d omai n; Fu rin-lik e cys teine rich region ; Gro wth f actor rece ptor dom ain IV		LANCL2 (NM_018697) - EGFR (NM_005228) rearrangement: c.1259-255:LANCL2_c.2947-344:EGFRdup Protein Fusion: out of frame {LANCL2:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059222-T01-IM7-540	P-0059222-T01-IM7		NIPSNAP2-EGFR	NIPSNAP2	unknown	three_prime	55233046	15	599	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region	Growth factor receptor domain IV	GBAS (NM_001483) - EGFR (NM_005228) fusion: c.92+717:GBAS_c.1796:EGFRdup Protein Fusion: mid-exon {GBAS:EGFR}	NA
EVT-P-0037759-T01-IM6-541	P-0037759-T01-IM6		NIPSNAP2-EGFR	NIPSNAP2	out-of-frame	three_prime	55268444	24	982	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		GBAS (NM_001483) - EGFR (NM_005228) Rearrangement: c.2946+338:EGFR_c.279-1159:GBASdup Protein Fusion: out of frame {GBAS:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037759-T03-IM6-542	P-0037759-T03-IM6		NIPSNAP2-EGFR	NIPSNAP2	out-of-frame	thre_prime	55268444	24	982	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Receptor L domain; Fusion-like cysteine rich region; Growth factor receptor domain IV		GBAS (NM_001483) - EGFR (NM_005228) rearrangement: c.279-1159:GBAS_c.2946+338:EGFRdup Protein Fusion: out of frame {GBAS:EGFR}	NA
EVT-P-0036028-T01-IM6-543	P-0036028-T01-IM6		NUMA1-EGFR	NUMA1	in-frame	thre_prime	55241128	17	687	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		NUMA1 (NM_006185) - EGFR (NM_005228) rearrangement: t(7;11)(p11.2;q13.4)(chr7:g.55241128::chr11:g.71728415) Protein Fusion: in frame {NUMA1:EGFR}	NA

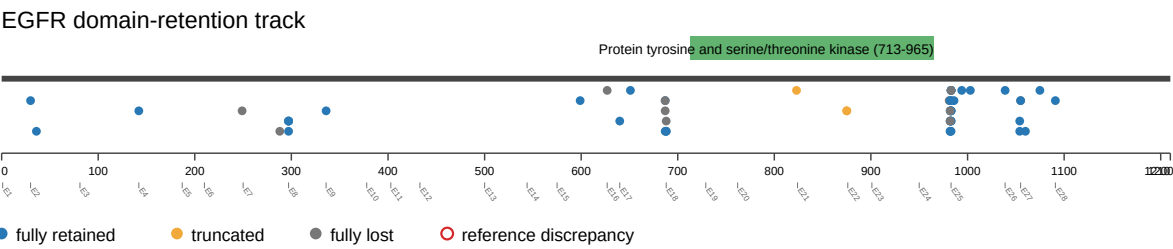
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 5470-T01-IM7-545	P-0065 470-T01-IM7		PDE1C-EGFR	PDE1C	unknown	threonine_prime	55240707	17	651	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		PDE1C (NM_001191058) - EGFR (NM_005228) fusion: c.86-15224:PDE1C_c.1951:EGFRinv Protein Fusion: mid-exon {PDE1C:EGFR}	NA
EVT-P-005 9032-T01-IM7-546	P-0059 032-T01-IM7		PLOD3-EGFR	PLOD3	out-of-frame	threonine_prime	55179084	2	30	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			PLOD3 (NM_001084) - EGFR (NM_005228) fusion: c.2062-9:PLOD3_c.89-30895:EGFRinv Protein Fusion: out of frame {PLOD3:EGFR}	NA
EVT-P-002 8484-T02-IM6-548	P-0028 484-T02-IM6		SCAF4-EGFR	SCAF4	out-of-frame	threonine_prime	55214145	4	142	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Furin-like cysteine rich region; Growth factor receptor domain IV		Receptor L domain	SCAF4 (NM_020706) - EGFR (NM_005228) fusion: t(7;21)(p11.2;q22.11)(chr7:g.55214145::chr21:g.33094864) Protein Fusion: out of frame {SCAF4:EGFR}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_truncated	retain_ed_domains	lost_domains	disrupt_ed_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 0910-T01-I M6-551	P-0030 910-T0 1-IM6		SEL1 L-EG FR	SEL 1L	in-fr ame	thre_e _pr im e	552224 85	7	297	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Grow th factor r ecepto r doma in IV	Rece ptor L do main	Furin-li ke cyst eine rich region	SEL1L (NM_005065) - EGFR (NM_005228) rearrangement: t(7;14)(p11.2 ;q31.1)(chr7:g.55222485::chr 14:g.81986576) Protein Fusion: in frame {SEL1L:EGFR}	NA
EVT-P-005 2976-T02-I M7-552	P-0052 976-T0 2-IM7		TNS3- EGFR	TNS 3	in-fr ame	thre_e _pr im e	552223 07	7	297	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Grow th factor r ecepto r doma in IV	Rece ptor L do main	Furin-li ke cyst eine rich region	TNS3 (NM_022748) - EGFR (NM_005228) fusion: c.1025- 2026:TNS3_c.889+462:EGF Rinv Protein Fusion: in frame {TNS3:EGFR}	NA
EVT-P-000 6873-T01-I M5-554	P-0006 873-T0 1-IM5		TUT7- EGFR	TUT 7	unk now n	five _pr im e	552380 31	16	627	True	lost	0.0	False	Recep tor L d omain; Furin-li ke cyst eine rich region	Prote in tyr osine and s erine /thre onine kinas e	Growth factor r eceptor domain IV	EGFR (NM_005228) - ZCCHC6 (NM_024617) rearrangement : t(7;9)(chr7 p11.2;chr9q21.33)(chr7:g.55 238031::chr9:g.88930920) Protein fusion: mid-exon (EGFR-ZCCHC6)	NA
EVT-P-001 5419-T01-I M6-555	P-0015 419-T0 1-IM6		VPS4 1-EG FR	VPS 41	unk now n	five _pr im e	552688 90	25	986	False	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Rece ptor L domai n; Furi n-like cystein e rich region; Growt h factor r ecepto r doma in IV			EGFR (NM_005228)- VPS41 (NM_014396) Rearrangement : c.2956:EG FR_c.247-4922:VPS41inv Protein fusion: mid-exon (EGFR-VPS41)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0030222-T01-IM6-557	P-0030222-T01-IM6		VWC2-EGFR	VWC2	out-of-frame	threonine_prime	55259621	21	875	True	disrupted	0.35968379446640314	True		Receptor domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase	VWC2 (NM_198570) - EGFR (NM_005228) rearrangement: c.827-37789: VWC2_c.2625+54:EGFRdel Protein Fusion: out of frame {VWC2:EGFR}	NA
EVT-P-0019922-T01-IM6-558	P-0019922-T01-IM6		YIF1B-EGFR	YIF1B	unknown	threonine_prime	55238939	16	640	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor domain; Furin-like cysteine rich region; Growth factor receptor domain IV		EGFR (NM_005228) rearrangement: t(7;19)(p22.1;q13.2)(chr7:g.5523893::chr19:g.38806683) Antisense Fusion	NA
EVT-P-0030222-T01-IM6-559	P-0030222-T01-IM6		ZBPB-EGFR	ZBPB	unknown	threonine_prime	55209996	2	36	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor domain; Furin-like cysteine rich region; Growth factor receptor domain IV			ZBPB (NM_007009) - EGFR (NM_005228) rearrangement: c.487+5719: ZBPB_c.106:EGFRinv Protein Fusion: mid-exon {ZBPB:EGFR}	NA

Figures

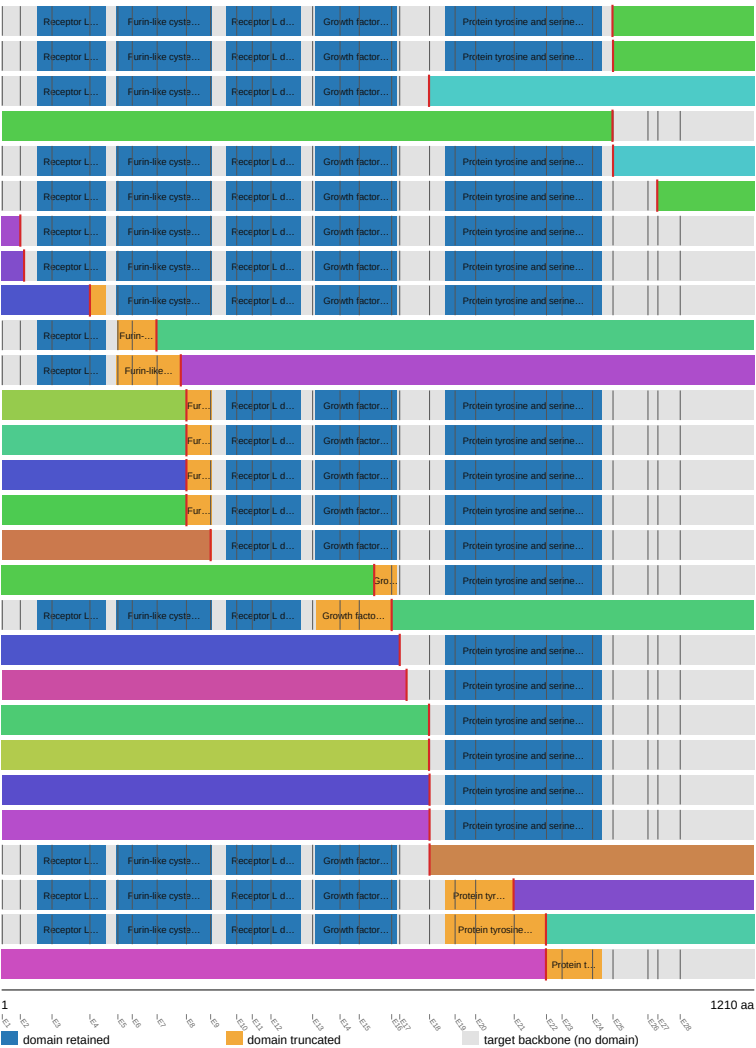
domain retention outliers



fusion schematic

EGFR fusion-transcript schematic

partner block → breakpoint → retained EGFR portion (domain-colored, exon ticks)



- SEPTIN14 (x6) – Protein tyrosine and serine/threonine kinase retained
- SEPTIN14 (x6) – Protein tyrosine and serine/threonine kinase retained
- FADD (x2) – Protein tyrosine and serine/threonine kinase lost
- NIPSNAP2 (x2) – Protein tyrosine and serine/threonine kinase lost
- RAD51 (x2) – Protein tyrosine and serine/threonine kinase retained
- SEPTIN14 (x2) – Protein tyrosine and serine/threonine kinase retained
- PLOD3 – Protein tyrosine and serine/threonine kinase retained
- ZBPB – Protein tyrosine and serine/threonine kinase retained
- SCAF4 – Protein tyrosine and serine/threonine kinase retained
- VOPP1 – Protein tyrosine and serine/threonine kinase lost
- PCDH15 – Protein tyrosine and serine/threonine kinase lost
- DDC – Protein tyrosine and serine/threonine kinase retained
- GARS1 – Protein tyrosine and serine/threonine kinase retained
- SEL1L – Protein tyrosine and serine/threonine kinase retained
- TNS3 – Protein tyrosine and serine/threonine kinase retained
- CSF2RA – Protein tyrosine and serine/threonine kinase retained
- NIPSNAP2 – Protein tyrosine and serine/threonine kinase retained
- TUT7 – Protein tyrosine and serine/threonine kinase lost
- YIF1B – Protein tyrosine and serine/threonine kinase retained
- PDE1C – Protein tyrosine and serine/threonine kinase retained
- CADPS – Protein tyrosine and serine/threonine kinase retained
- NUMA1 – Protein tyrosine and serine/threonine kinase retained
- EEA1 – Protein tyrosine and serine/threonine kinase retained
- KIF5B – Protein tyrosine and serine/threonine kinase retained
- VSTM2A – Protein tyrosine and serine/threonine kinase lost
- CDH7 – Protein tyrosine and serine/threonine kinase truncated
- LANCL2 – Protein tyrosine and serine/threonine kinase truncated
- VWC2 – Protein tyrosine and serine/threonine kinase truncated

Showing the top 28 of 41 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic

EGFR intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



reference comparison

Reference vs msk_impact_50k_2026

